

Thymeleaf Cheat Sheet

Setup and properties

Deps spring-boot-starter-thymeleaf + spring-boot-starter-webmvc (or -webflux); thymeleaf-extras-springsecurity6;

nz.net.ultraq.thymeleaf:thymeleaf-layout-dialect (Boot-managed); htmx-spring-boot-thymeleaf 5.2.0 (not in BOM). No thymeleaf-spring7: Spring 7 uses thymeleaf-spring6.

Locations src/main/resources/templates/ (.html), static/, messages*.properties. Modes: HTML, XML, TEXT, JAVASCRIPT, CSS, RAW.

spring.thymeleaf.cache	true
.prefix / .suffix	classpath:/templates/ / .html
.mode / .encoding	HTML / UTF-8
.check-template / .check-template-location	true / true
.enable-spring-el-compiler	false
.view-names .excluded-view-names .template-resolver-order	filters
.servlet.content-type	text/html
.reactive.max-chunk-size	0B
.reactive.*-mode-view-names	

spring.thymeleaf.enabled is deprecated (error level). Extra resolvers/dialects: declare ITemplateResolver / IDialect beans. DevTools turns the cache off.

Expressions and basic objects

<code>\${x}</code>	variable (OGNL; SpEL with Spring)
<code>*{x}</code>	selection on th:object
<code>#{key(a,b)}</code>	message (i18n)
<code>@{/p/{id}{id=\${i}}}</code>	link URL
<code>~{tpl::sel}</code>	fragment expression

Objects #ctx (use this), #vars/#root, #locale; namespaces \${param.q[0]}, \${session.x}, \${application.x}; request data via #ctx.getExchange(). Safe navigation \${a?.b}; beans \${@bean.m()} (SpEL); T(java.time.Year) (blocked in restricted mode). Selection falls back to \${#object}.

Literals, operators, preprocessing

Literals	'text' 42 3.1 true null tokens one
Text	'a' + \${b}, substitution Hi \${n}
Arithmetic	+ - * / % (div mod), unary -
Compare	> < >= <= (gt lt ge le), == != (eq ne)
Boolean	and or, ! not
Conditional	(c) ? a : b, (c) ? a, Elvis x ?: d
No-op	_ keeps the prototype text
Conversion	\${{x}} / *{{x}} via ConversionService
Preprocess	__\${x}__ evaluated first (never with user input)

Utility objects

#execInfo	template name and mode
#messages	msg, msgOrNull, arrayMsg
#uris	escapePath, escapeQueryParam
#conversions	convert(obj, 'String')
#dates #calendars	format, year, createNow
#temporals	java.time format (merged in 3.1)
#numbers	formatDecimal, formatCurrency, sequence
#strings	isEmpty, contains, abbreviate, capitalize, join
#objects #bools	nullSafe, isTrue
#arrays #lists	size, isEmpty, contains, sort
#sets #maps	
#aggregates	sum, avg
#ids	seq, next, prev
Spring	#fields, #mvc.url; security: #authentication, #authorization

Security strip

th:text and [...] escape for HTML; th:utext and [...] do not: sanitize first (OWASP Java HTML Sanitizer). JS inlining escapes for JS. Never put user input in __\${x}__ preprocessing, view names, fragment selectors or th:attr names. Restricted mode blocks T(), new, #execInfo, \${param} in th:on*. Use 3.1.5+: CVE-2026-41901, -40478, -40477 (sandbox bypasses), CVE-2021-43466, CVE-2023-38286. Add a CSP nonce, keep CSRF on.

th:* attributes by purpose

Text	th:text (escaped), th:utext (raw)
Attributes	th:attr="a=\${x}", th:value th:href th:src th:action, th:classappend th:styleappend th:attrappend th:attrprepend, booleans th:checked th:disabled th:selected th:readonly; th:data=*, th:hx=*, th:aria=* by default processor
Control	th:each, th:if / th:unless, th:switch / th:case="*"
Variables	th:with="a=\${x},b=2", th:object
Fragments	th:fragment th:insert th:replace th:remove th:assert th:ref th:block
Other	th:inline, th:field th:errors th:errorclass, th:alt= title, th:lang= xml:lang

Precedence (low number runs first):

- fragment inclusion: th:insert th:replace
- iteration: th:each
- condition: th:if th:unless th:switch th:case
- local vars: th:object th:with
- general attrs: th:attr th:attrprepend th:attrappend
- specific attrs: th:value th:href th:src ...
- text: th:text th:utext
- fragment spec: th:fragment
- removal: th:remove

A th:with variable is not visible to th:if/th:each on the same tag.

Iteration status and truthiness

th:each="b, s : \${books}" (status var defaults to bStat): s.index (from 0) s.count (from 1) s.size s.current s.even s.odd (first row is odd) s.first s.last. Iterates Iterable, Map (entries), arrays, Iterator, Enumeration, Stream (once, no side).

LazyContextVariable defers loading.

True: boolean true; non-zero number or char; any string except 'false', 'off', 'no'; any other non-null object.

False: null, false, 0, those three strings. th:unless inverts. Empty string is true.

Fragments and the Layout Dialect

th:fragment="card(book,css)"; th:insert=~{frag/card :: card(\${b},'x')} keeps the host tag, th:replace swaps it. Selectors: ~{tpl} whole, ~{tpl :: name}, ~{:: name} / ~{this :: name}, CSS-like markup selectors #id .class div[0] (0-based). Named params card(book=\${b}); markup args ~{::#row}; ~{} empty; _ no-op; th:assert checks; th:include deprecated; th:substituteby removed.

th:remove all | body | tag | all-but-first | none.

Layout Dialect 4.0.1 xmlns:layout: layout:decorate=~{layout/main}" (page), layout:fragment="content" (in both), layout:title-pattern="LAYOUT_TITLE - \$CONTENT_TITLE", layout:insert / layout:replace; head elements merged (sorting strategy configurable); layout:collect removed in 4.0.

Comments, blocks and inlining

<!--/* note */--> parser-level, removed. <!--/*/ <th:block th:each="x : \${l}"> /*/--> prototype-only markup. Standard <!-- --> is output (and inlined). <th:block> = attribute container, tag vanishes.

Inline [[\${x}]] escaped, [\${x}] unescaped; th:inline="none" disables; JS: var b = /* [[\${book}]]*/ null; (JSON, escaped), CSS /* [[\${c}]]*/. TEXT/JS/CSS modes: [# th:each="x : \${l}"]...[/], [# th:if="\${c}"], [#th:block]; /*[- ... -]*/ and /*[+ ... +]*/ prototype/engine-only comments. Decoupled logic: .th.xml next to the HTML.

Changed since older tutorials

#request #session #servletContext removed in 3.1 (use param, session, application, #ctx.getExchange()) · th:include deprecated, th:substituteby removed · thymeleaf-extras-java8time merged into core (#temporals) · springsecurity5 → 6, javax → jakarta · Layout Dialect layout:decorator → layout:decorate · WebSecurityConfigurerAdapter removed (use SecurityFilterChain) · htmx 1 → 2: extensions are separate packages, DELETE params in URL · @MockBean → @MockitoBean · DevTools LiveReload deprecated · server.error.* → spring.web.error.*.

Spring forms and validation

<form th:action="@{/books}" th:object="\${bookForm}" method="post"> (CSRF hidden input added when Spring Security is on). th:field="*{title}" sets id, name, value: text/number/date/email/password inputs, textarea, select (+ th:each options, th:value), checkbox (boolean, or collection with th:value; hidden _name marker), radio. method="put" + _method filter.

Errors controller: @Valid Form f, BindingResult r (BindingResult right after the form). Template: #fields.hasErrors('title') / ('*') / ('all') / ('global'), #fields.errors(), #fields.detailedErrors(), th:errors="*{title}", th:errorclass="is-invalid". Redirect after POST; flash attributes survive one redirect.

i18n, conversion, links

#{home.title}, #{greet(\${u.name})}, dynamic #{|status.\${s}|}; spring.messages.basename; LocaleChangeInterceptor + resolver (?lang=es); missing key renders ??key_es??. Formatting: \${{d}}, #temporals.format, @DateTimeFormat, @NumberFormat, custom Formatter.

Links @{/books} context-relative; @~/other} server-relative; @/cdn/x} protocol-relative; @{https://h/p} absolute; @{book.html} page-relative; @{/books/{id} (id=\${b.id}, tab='x')} path variable + query; @{/s(q=\${q}, tag=\${t})}; @{/p#frag}; \${#mvc.url('BC#show').arg(0,id).build()}; static versions via ResourceUrlEncodingFilter.

Spring Security

sec.authorize="hasRole('LIBRARIAN')", sec.authorize="isAuthenticated()", sec.authorize-expr="\${perm}", sec.authorize-url="POST /books", sec.authorize-acl; sec.authentication="name", sec.authentication="principal.authorities"; objects \${#authentication.name}, \${#authorization.expression('hasRole(''ADMIN'')')}. Hiding links is not authorization: enforce with authorizeHttpRequests and @PreAuthorize. Logout is a POST with CSRF; springsecurity5 became springsecurity6.

Fragment rendering and htmx

Fragments return "books/list :: rows" or FragmentsRendering.with(...) (several fragments, OOB).

Attributes th:hx-get="@{/books}" (hx-get/post/put/patch/delete), hx-target hx-swap (innerHTML outerHTML before/after delete) hx-trigger (keyup changed delay:300ms, revealed) hx-select hx-push-url hx-boost hx-indicator hx-vals hx-confirm hx-swap-oob hx-on.

Headers in HX-Request HX-Target HX-Trigger HX-Boosted HX-Current-URL. **out** HX-Redirect HX-Refresh HX-Retarget HX-Reswap HX-Trigger HX-Push-Url HX-Location.

htmx-spring-boot 5.2.0 @HxRequest @HxGetMapping ..., HtmxRequest, HtmxResponse, @HxTrigger @HxRetarget @HxReswap @HxPushUrl @HxRefresh @HxRedirect, HtmxRedirectView, redirect:htmx; hx: dialect. Send the CSRF header; SSE and WS via 2.x extensions.

Reactive, testing, extending

WebFlux SpringWebFluxTemplateEngine, ThymeleafReactiveViewResolver; modes FULL, CHUNKED, DATA-DRIVEN; new ReactiveDataDriverContextVariable(flux, 1); one data driver per template; back-pressure by buffer size.

Testing @WebMvcTest + MockMvcTester, @MockitoBean, security test support, HtmlUnit, Playwright; thymeleaf-testing last release 3.0.4 (3.1.0.M2 only). Boot 4: spring-boot-starter-webmvc-test.

Extending IDialect / AbstractProcessorDialect (+ IExpressionObjectDialect, IPreProcessorDialect, IPostProcessorDialect); processors: attribute, element, model, comment, CDATA, text, doctype, XML declaration, processing instruction, template boundaries; dialect prefix, precedence, IModelFactory.